



Research article

A multi-chamber incubation system for accurate assessment of ammonia and hydrogen sulfide emissions from cattle manure: Effects of dietary protein level

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ABSTRACT

A simplified multi-chamber manure (mixture of feces and urine) incubation system was developed to accurately measure ammonia (NH₃) and hydrogen sulfide (H₂S) emissions from manure. Addressing the complexity of conventional equipment, this methodological innovation replaces individual mass flow controllers and real-time gas analyzers with a practical 24-hour composite sampling approach. The system comprises six chambers with controlled airflow (2 L/minute) and humidity, directing vented air into Tedlar bags for reliable daily quantification. Accuracy was evaluated by introducing a known concentration of NH₃ (92 ppm) into empty chambers, resulting in a mean recovery rate of 95.7% with no significant differences among chambers. Precision was assessed by incubating identical manure samples in all chambers over five days, demonstrating consistent NH₃ emission rates (coefficient of variation: 2.6%). As an application example, the validated system was used to evaluate the effects of dietary crude protein (CP) levels on NH₃ and H₂S emissions from manure of Hanwoo steers during a 10-day incubation. The system successfully captured emission dynamics, clearly detecting that a high-CP diet significantly increased daily NH₃ emission rates ($P < 0.001$) compared to lower-CP diets, while confirming that cumulative NH₃ and H₂S emissions remained unaffected by the dietary CP levels. Ultimately, this cost-effective and standardized system bridges the gap between complex laboratory instrumentation and practical nutritional research, providing a robust platform for evaluating dietary strategies aimed at mitigating livestock odor emissions.

1. Introduction

As the global population has increased, the demand for livestock products has also increased. Global meat consumption increased from approximately 200 million tons in 2010 to about 300 million tons in 2018 (Parlasca and Qaim, 2022). In turn, the increased

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number of livestock has resulted in a significant increase in the amount of manure being excreted, and that can lead to environmental pollution. Animal manure contains nitrogen and sulfur, which can be converted by microbial activity into volatile gases such as ammonia (NH_3) and hydrogen sulfide (H_2S) (Font-Palma, 2019; Hu et al., 2022; Meng et al., 2023).

Both NH_3 and H_2S contribute to air pollution, odor nuisance, and contamination of soil and water (NRC (National Research Council), 2003). NH_3 is of particular concern because it can be oxidized to nitrous oxide (N_2O), a potent greenhouse gas with a 100-year global warming potential approximately 300 times greater than that of carbon dioxide (Amaral Junior et al., 2024; Ju et al., 2025; Paudel et al., 2019), and it also contributes to ozone layer depletion (Ravishankara et al., 2009). H_2S , on the other hand, poses serious health risks; chronic exposure to concentrations as low as 10 ppm can be harmful, and acute exposure above 500 ppm can be life-threatening (Costigan, 2003; Rabin et al., 2025). If emissions from livestock manure are not effectively managed, they will exacerbate odor problems, greenhouse gas release, and environmental degradation (Cao et al., 2023; Wang et al., 2017). In response to these concerns, recent studies have examined feeding strategies, such as reducing dietary crude protein (CP) content with amino acid supplementation or adding digestive enzymes, microbial agents, and probiotics, to mitigate emissions while maintaining productivity (Lee et al., 2021; Pan et al., 2017; Shriver et al., 2003).

Assessment of such strategies requires accurate and continuous measurement of manure gas emissions. Historically, manure gas concentrations have been measured using single readings with colorimetric tubes, acid traps, and electrochemical sensors (Paulas and Lemort, 2024; Wheeler et al., 2011). Recent studies have explored indirect measurement approaches using biomarkers to estimate NH_3 emissions from manure, providing complementary methods to direct gas quantification (Hristov et al., 2009; Song et al., 2024). However, microbial activity during manure fermentation dynamically alters gas flux patterns over time, meaning single readings or indirect estimations are often insufficient. Furthermore, controlling one target gas can unintentionally increase emissions of another—a phenomenon known as "pollution swapping" (e.g., lowering NH_3 while increasing H_2S or N_2O ; Newell Price et al., 2011), necessitating continuous multi-gas monitoring.

To evaluate these dynamic emissions continuously, chamber methodologies are widely used, generally classified into static and dynamic systems. While static chambers are relatively simple, they suffer from gas accumulation that can artificially suppress emission rates and alter biological processes (Heinemeyer and McNamara, 2011). Conversely, dynamic flow-through chambers better simulate field conditions by continuously supplying fresh air, though they require precise airflow regulation. To address this, Wheeler et al. (2011) developed an automated dynamic incubation system. However, its heavy reliance on highly expensive components, specifically a photoacoustic multi-gas monitor and individual mass flow controllers (MFCs) for continuous real-time air routing, makes it complex and cost-prohibitive.

To address these limitations, we developed a simplified multi-chamber manure incubation system designed to quantitatively measure NH_3 and H_2S emissions with high precision and reliability under controlled airflow and humidity using a more practical sampling approach. In this study, we describe the design and operation of this system, evaluate its accuracy and precision, and demonstrate its application in comparing manure emissions from Hanwoo steers fed diets with varying CP levels.

2. Materials and methods

2.1. Development of a manure incubation system

A continuous air flux multi-chamber system was constructed, adopting the core incubation principles described by Wheeler et al. (2011). In order to enhance practicality of the system and reduce its technical complexity, a significant simplification was made,

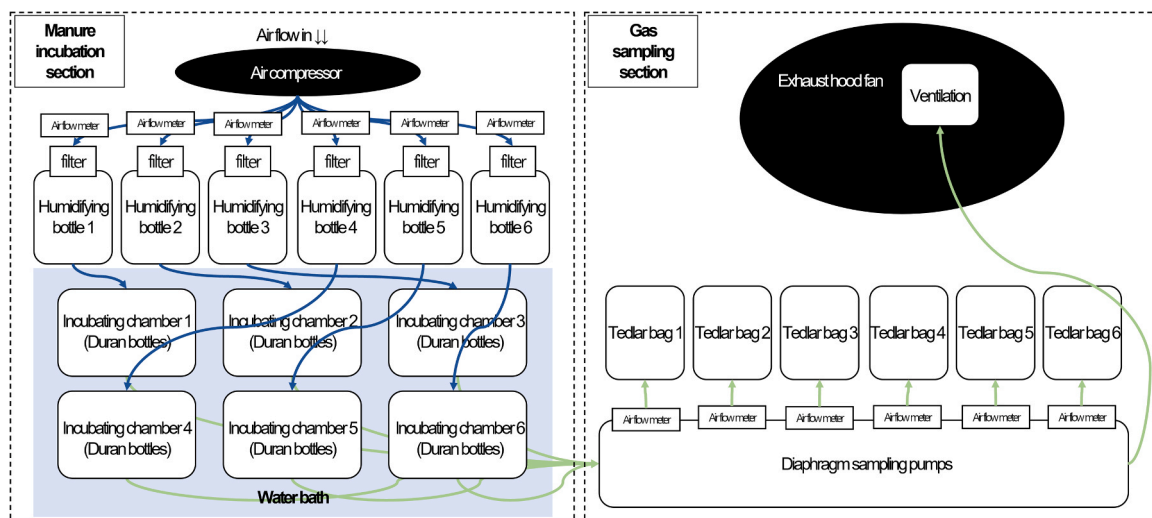


Fig. 1. Schematic diagram of the incubation system developed in this study.

replacing the individual MFCs and complex real-time analyzer with a 24-hour composite sampling approach. This modification employs a timer-controlled diaphragm pump to collect exhaust air into Tedlar bags at 30-minute intervals, enabling reliable daily emission quantification without the requirement for intricate flow regulation hardware. The chamber design prioritized controlled airflow and humidity to reflect emission dynamics across manure handling stages (Mohankumar Sajeev et al., 2018). The system was divided into two main parts: the manure incubation section and the gas sampling section (Fig. 1 and Supplementary Material 2). In this study, manure refers to the mixture of feces and urine at the time of incubation.

2.1.1. Manure incubation section

The manure incubation section consisted of incubation chambers, humidifying bottles, an air compressor, an airflow meter, and a circulating water bath. The system (overall dimensions = 840 mm × 540 mm × 635 mm, W × D × H) consisted of six chambers (1.0 L Duran bottles) equipped with lids that provide an inlet and outlet for air. Each chamber was connected to a humidifying bottle (500 mL Duran bottle) attached to the air inlet and to the gas sampling section via the air outlet. An air compressor (OFSA800–24, COMP WORLD, Gwangju, Korea) provided fresh air at a flow rate of 2.0 L/min per chamber using an airflow meter (RMA-0(0.1–1 L)-SSV, Dwyer, Michigan City, USA) and passed through a 0.22 µm PTFE filter (filtraTeCH, Orleans, France) into the humidifying bottles. This flow rate was selected based on previous work indicating that it was typical of air exchange rates in naturally ventilated livestock facilities and was suitable for gas emission measurements from manure (Wheeler et al., 2011). The outflow of moistened air from the humidifying bottles prevented drying of the manure surface. Each humidifying bottle contained 500 mL of water, which was determined to be the maximum amount required to maintain consistent humidity and prevent surface drying for up to 10 days during incubation. The temperature inside the chambers was maintained at room temperature (19°C) using a circulating water bath with a temperature range of –20°C to 100°C (± 0.1°C) (MaXircu™ CR-8, DaiHan Scientific Co., Ltd., Daejeon, Korea).

2.1.2. Gas sampling section

The outlet from each chamber was connected to an exhaust hood fan (SSFH 2012, ShinSaeng Scientific Co., Ltd., Paju, Korea) using Teflon-lined PVC tubing (0.32-mm i.d.; 0.64-mm o.d.; Tygon SE-200 Saint Gobain, Akron, OH) with a sampling line. This line was branched from the exhaust line and connected to a diaphragm sampling pump (KPM27H; Koge Electronics Co. Ltd., Taipei, Taiwan). The operation of the pump was controlled using an automatic sampling system consisting of 150 W enclosed switching power supply (LRS-150–12, MEAN WELL Enterprises Co., Ltd., Taipei), a miniature power relay (KH-103–2CL, KOINO, Seoul, Korea), and a timer (LE4S, Autronics, Busan, Korea). The pump collected the exhaust air from the chamber into a 20 L Tedlar bag at 30-minute intervals over a 24-hour sampling period. The air from the compressor to the Tedlar bag and hood flowed through the tubing throughout the system. At the end of each 24-hour sampling period, the collected air samples analyzed for NH₃ using a photoacoustic gas analyzer (LGD compact-A NH₃, Axetris AG, Kaegiswil, Switzerland, with detection range of 0–100 ppm NH₃).

2.2. System performance evaluation

2.2.1. Accuracy test

To evaluate the performance of the system, accuracy tests were conducted following the recovery rate validation protocol described by Wheeler et al. (2011), with modifications for multi-chamber consistency. The system was evaluated to determine whether the same concentration result could be obtained when collecting gas at a known concentration of NH₃. NH₃ gas (92 ppm) was introduced into the empty incubation chambers at a flow rate of 2.0 L/minute. The accuracy test was conducted using a replicate-in-time approach with three replicates. The recovery rate (%) was calculated using Eq.1. The dilution effect caused by the volume of air already present (30 L) in the chamber before the NH₃ gas is introduced (32 L) was considered.

$$\text{Recovery rate (\%)} = \frac{\text{NH}_3 \text{ concentration in the outflow air (ppm)}}{\text{NH}_3 \text{ concentration in the inflow air (92 ppm)}} \times \frac{\text{Pre-existing air (30 L)}}{\text{Chamber volume (32 L)}} \times 100, \quad (1)$$

2.2.2. Precision test

A precision test was also conducted to assess the consistency and repeatability of NH₃ measurements in each chamber over five days. The test protocol was adapted from standardized incubation approaches (Ellersiek and Olf, 2024). For the precision test, a homogeneous manure sample was prepared by collecting feces and urine from steers fed four kilograms of tall fescue supplemented with four kilograms of a conventional concentrate mix, mixed at a urine-to-feces ratio of 1:1 (w/w). The manure was subsampled with 320 g and incubated for five days at each chamber. A 1:1 (w/w) urine-to-feces ratio was chosen as it was considered the highest NH₃ release potential under controlled experimental conditions (Wheeler et al., 2011). The amount of manure placed in the chambers was selected based on preliminary experiments. The gas vented from the incubation chambers was collected every 30 min to measure the concentration of NH₃ at 24-hour intervals for five days. To calculate the production rate of mg NH₃/g manure/hour, the final value was divided by the 320 g of manure incubated [Eq.2].

$$\text{NH}_3 \text{ production rate (mg NH}_3\text{/g manure/hour)} = \text{NH}_3 \text{ concentration (ppm)} \times (10^{-6}\text{L/L})/\text{ppm} \times \text{air flow (L/minute)} \times \text{density (0.769 g of NH}_3\text{/L at standard temperature and pressure)} \times 60 \text{ min/hour} \div 320 \text{ g of manure} \quad (2)$$

2.3. Overview of the experimental procedure

Manure collected from a feeding trial was incubated using the developed multi-chamber system to compare potential NH_3 and H_2S emissions resulting from diets with different crude protein (CP) levels. The feeding trial involved four 17-month-old Hanwoo steers (477 ± 2.9 kg) in the early-fattening period, arranged in a 4×4 cross-over design with four dietary treatments across four periods. Treatment sequences were randomized and balanced to minimize carry-over effects (Seo et al., 2018). Each period lasted 21 days: a 14-day adaptation phase followed by a 7-day sampling phase, with a 7-day wash-out interval between periods. The dietary treatments consisted of concentrate mixes containing 12% (LCP), 15.5% (MLCP), 17.25% (MHCP), or 19% (HCP) on an as-fed basis. The concentrate mixes were primarily composed of flaked corn, ground corn, ground wheat, corn gluten feed, soybean hull, wheat bran, dried distillers grains with soluble, and palm kernel meal (Supplementary Table 1) and were formulated to differ in CP content while maintaining similar dietary energy levels (Table 1).

The steers were housed individually in $5 \text{ m} \times 5 \text{ m}$ pens equipped with forage feed bunks, enabling automatic measurement of individual forage intake via radio-frequency identification tags (Dawoon Co., Incheon, Korea). Feeding was carried out twice daily at 08:00 and 18:00 h. Tall fescue hay was provided as forage, and water was available *ad libitum*. Each steer received an equal allotment of concentrate mix, targeting an average daily gain of 0.9 kg/day according to the Korean feeding standards for Hanwoo growing steers (NIAS National Institute of Animal Science, 2017). Dry matter intake (DMI) was recorded daily; forage intake was automatically logged, while concentrate intake was calculated by measuring orts after a 30-minute feeding period.

Table 1

Analyzed chemical composition (g/kg DM or as stated) of the experimental diets.

Items ^a	Treatment ^b				Tall fescue
	LCP	MLCP	MHCP	HCP	
DM, g/kg as fed	886	886	886	886	888
OM	911	916	918	920	941
CP	147	178	193	208	69
SOLP	58	67	72	77	30
NDICP	26	27	27	27	13
ADICP	12	13	14	15	9
aNDF	271	286	293	301	640
ADF	134	131	129	128	404
ADL	38	37	36	36	62
Ether extract	69	62	59	56	13
Ash	89	84	82	80	59
Ca	15	13	12	10	3
P	6	6	7	7	1
K	10	11	11	12	18
Na	2	2	3	3	1
Cl	5	5	5	5	5
S	4	4	5	5	1
Mg	3	4	4	4	1
TDN	747	740	736	733	557
NEm, MJ/kg DM	8.1	8.0	7.9	7.8	5.0
NEg, MJ/kg DM	5.4	5.3	5.2	5.2	2.7
Total carbohydrates	695	676	666	656	859
NFC	450	417	400	383	232
Carbohydrate fraction, g/kg carbohydrate					
CA	84	88	91	93	127
CB1	554	506	481	457	2
CB2	9	21	27	34	141
CB3	221	253	269	285	556
CC	132	131	131	131	174
Protein fraction, g/kg CP					
PA+B1	391	381	376	371	435
PB2	431	463	480	496	374
PB3	98	80	70	61	64
PC	79	76	74	72	128

^a DM: dry matter, OM: organic matter, CP: crude protein, SOLP: soluble CP, NDICP: neutral detergent insoluble CP, ADICP: acid detergent insoluble CP, aNDF: neutral detergent fiber analyzed using a heat stable amylase and expressed inclusive of residual ash, ADF: acid detergent fiber, ADL: acid detergent lignin, TDN: total digestible nutrients, NEm: net energy for maintenance, NEg: net energy for growth, NFC: non-fiber carbohydrate, CA: carbohydrate A fraction; ethanol soluble carbohydrates, CB1: carbohydrate B1 fraction; starch, CB2: carbohydrate B2 fraction; soluble fiber, CB3: carbohydrate B3 fraction; available insoluble fiber, CC: carbohydrate C fraction; unavailable carbohydrate, PA+B1: protein A and B1 fractions; soluble CP, PB2: protein B2 fraction; intermediate degradable CP, PB3: protein B3 fraction; slowly degradable fiber-bound CP, PC: protein C fraction; unavailable CP.

^b LCP, Low crude protein (12%); MLCP, Middle-low crude protein (15.5%); MHCP, Middle-high crude protein (17.25%); HCP, High crude protein (19%)

2.4. Manure collection

Feces and urine were collected from all steers via spot sampling during the final week of each experimental period. Sampling was conducted once every nine hours over four consecutive days, resulting in eight fecal and eight urinary samples per steer. Sampling times were distributed to represent all hours of the day within a 24-hour cycle. For example, samples were collected at 17:00 h on day 1; at 02:00, 11:00, and 20:00 h on day 2; at 05:00, 14:00, and 23:00 h on day 3; and at 08:00 h on day 4.

At each fecal sampling, approximately 300 g of feces were obtained directly from the rectum. Samples were thoroughly homogenized, and approximately 200 g was transferred to an aluminum tray and dried at 55 °C in a forced-air oven for 72 h to determine dry matter (DM) content. The remaining approximately 100 g of fresh feces was immediately frozen at −20 °C for subsequent manure incubation. The dried fecal samples from each steer were later composited on an equal dry-weight basis for chemical analysis.

At each urine sampling, approximately 150 mL was collected using a container positioned beneath the steer's abdomen. The urine was filtered through four layers of cheesecloth into a Duran bottle and subsequently sub-sampled into three 50 mL Falcon tubes. All subsamples were immediately frozen at −20 °C. For nitrogen content determination, one subsample was acidified with 2 M H₂SO₄ prior to freezing.

2.5. Chemical analysis of feeds and manures

The forage, concentrate mix, and dried fecal samples were ground using a cyclone mill (Foss, Hillerød, Denmark) fitted with a 1-mm screen prior to chemical analysis. Chemical composition was determined by Cumberland Valley Analytical Services, Inc. (Hagerstown, MD, USA) using procedures described in Jeon et al. (2016). DM was determined as total solids, and total volatile solids (VS; organic matter, OM) were calculated as DM minus ash content. Total nitrogen (N) was measured by combustion using a nitrogen combustion analyzer (Leco FP-528, Leco, MI, USA). Fecal samples were also analyzed for acid-insoluble ash (AIA) according to Van Soest et al. (1991) to estimate daily fecal output for each animal. Total sulfur (S) concentration in fecal samples was determined by inductively coupled plasma emission spectroscopy (ICP; IRIS Interpid II XDL, Thermo Fisher Scientific Inc., Waltham, MA, USA) following microwave digestion with nitric acid (Qwave 2000, Questron Technologies Corp., Mississauga, Canada) as described by Isaac and Johnson (1985).

Urine samples were thawed and composited in equal volumes for analysis. Total N in acidified urine was measured by the Kjeldahl method using a DK 20 Heating Digester and a semi-automatic distillation unit (UDK 139, VELP Scientifica, Usmate, Italy). Urinary total S concentration was determined by ICP as described above. Urinary creatinine concentrations were measured using a commercial assay kit (No. 500701; Cayman Chemical, Ann Arbor, MI, USA) and were used to estimate each animal's daily urine output according to Tebbe and Weiss (2018), assuming a creatinine excretion rate of 29.0 mg/kg body weight based on Valadares et al. (1999).

2.6. Manure incubation

For each incubation batch, fecal samples from all animals within the same experimental period were pooled to eliminate potential variability among incubation batches that could influence treatment effects. Fecal and urinary samples were thawed at 2 °C for two days prior to incubation to minimize dripping. After thawing, feces and urine were combined to prepare a 320 g manure sample for each animal, with mixing ratios determined according to each steer's average daily fecal and urinary output. Each manure sample was placed in a separate chamber and incubated at 20 °C for 10 days. The inflow air rate was maintained at 2 L/minute, and the outflow was directed to a 20 L Tedlar bag via a gas sampling device connected to the outflow line. Outflow sampling was conducted at 0.4 L/minute for 30 s every 30 min over a 24-hour cycle. Sampling bags were replaced daily. One empty chamber served as a blank to correct for background NH₃ and H₂S levels.

Air samples collected over each 24-hour period were analyzed for NH₃ and H₂S using a photoacoustic gas analyzer (NH₃: LGD compact-A NH₃, Axetris AG, Kaegiswil, Switzerland; detection range 0–100 ppm; H₂S: C-50, Membrapor AG, Wallisellen, Switzerland; detection range 0 ppb to 50 ppm). Concentrations measured in the daily samples were converted to mass emissions following Lee et al. (2020). Briefly, corrected daily concentrations of NH₃ (ppm) and H₂S (ppb) were multiplied by the daily air flux (2 L/minute), 60 min/hour, and each gas's density at standard temperature (0 °C) and pressure (760 Torr: NH₃ = 0.769 g/L; H₂S = 1.505 g/L) to obtain mass. Emissions were expressed per gram of manure by dividing by the 320 g incubated (Eqs. 1 and 2). Daily emissions (Eqs. 5 and 6) and cumulative 10-day emissions (Eqs. 7 and 8) were calculated by summing daily values, where *i* represents the day of incubation.

$$\text{mg NH}_3/\text{g manure/hour (A)} = [\text{NH}_3 \text{ concentration (ppm)} - \text{blank NH}_3 \text{ concentration (ppm)}] \times (10^{-6} \text{ L/L})/\text{ppm} \times \text{air flow (L/minute)} \times \text{density (0.769 g of NH}_3/\text{L at standard temperature and pressure)} \times 60 \text{ min/hour} \div 320 \text{ g of manure} \quad (3)$$

$$\mu\text{g H}_2\text{S/g manure/per hour (B)} = [\text{H}_2\text{S concentration (ppb)} - \text{blank H}_2\text{S concentration (ppb)}] \times (10^{-6} \text{ L/L})/\text{ppm} \times \text{air flow (L/minute)} \times \text{density (1.505 g of H}_2\text{S /L at standard temperature and pressure)} \times 60 \text{ min/hour} \div 320 \text{ g of manure} \quad (4)$$

$$\text{g NH}_3/\text{day/steer (C)} = \text{A} \times 24 \text{ h/day} \times \text{estimated daily total manure output (kg/day/steer)} \quad (5)$$

$$\text{mg H}_2\text{S/day/steer (D)} = \text{B} \times 24 \text{ h/day} \times \text{estimated daily total manure output (kg/day/steer)} \quad (6)$$

$$\text{CumulativeNH}_3(\text{g} / \text{steer over 10 days}) = \sum_{i=1}^{10} C \quad (7)$$

$$\text{CumulativeH}_2\text{S}(\text{g} / \text{steer over 10 days}) = \sum_{i=1}^{10} D \quad (8)$$

2.7. Approval of animal experiments and statistical analysis

The animals were treated in accordance with the animal experimental protocol approved by the Institutional Animal Care and Use Committee (IACUC) of Chungnam National University (Approval No. 202304A-CNU-005) for feeding and manure collection.

Data management and statistical analyses were performed using Microsoft Excel (Microsoft Corp., Redmond, WA, USA), SPSS version 26 (IBM Corp., Armonk, NY, USA), and R version 4.3.1 (R Core Team, Vienna, Austria). Excel was used to organize raw data, calculate descriptive statistics (means and standard deviations), and prepare formatted datasets for statistical analysis. Manure characterization data, incubation system performance results, and manure incubation measurements were analyzed using one-way analysis of variance (ANOVA) with a significance threshold of $P < 0.05$ in SPSS. Tukey's post-hoc test was applied to identify pairwise differences between treatments. For the manure incubation results, least-squares means (LS-means) were additionally calculated, and polynomial contrasts (linear and quadratic) were tested to evaluate trends associated with incremental treatment levels.

3. Results and discussion

3.1. System performance evaluation

3.1.1. Accuracy test

The accuracy of the newly developed manure incubation system is described in Table 2. When a known concentration (92 ppm) of NH_3 gas was introduced into the system, the mean concentrations of NH_3 in the sampled air from chambers 1, 2, 3, 4, 5, and 6 were 81.59 ± 0.79 ppm, 82.89 ± 0.99 ppm, 81.73 ± 1.14 ppm, 83.83 ± 1.20 ppm, 82.38 ± 0.33 ppm, and 82.60 ± 0.19 ppm, respectively. There were no significant differences among chambers ($P = 0.188$), indicating consistent system performance. The inlet air had an NH_3 concentration of 92 ppm, while the collected air had a mean concentration of 82.50 ± 1.18 ppm (mean $\pm$ SD). This discrepancy may be due to the mixing of incoming air with the pre-existing air within the chambers, resulting in a diluted concentration of NH_3 . Considering the pre-existing air in the system, the estimated actual concentration of NH_3 was 86.25 ppm, resulting in a mean recovery rate of $95.7 \pm 1.4\%$ (mean $\pm$ SD) according to [Eq.1]. The discrepancy, approximately 3.5 ppm might be attributed to the absorption of NH_3 onto the tubes or bottles during the process (Wheeler et al., 2011).

3.1.2. Precision test

The results from the precision test also indicated consistent values of NH_3 emissions among the six chambers (Table 3). NH_3 production over five days averaged 0.0038 ± 0.0003 (mg NH_3 /g manure/hour) and did not differ significantly ($P = 0.082$) among one-way ANOVA with repeated measures. With the high recovery rate (95.7%) and a low coefficient of variance (2.6%) for NH_3 production rate, it can be concluded that our incubation system can be used to reliably test differences in gas emissions between samples.

3.1.3. Methodological advancements and limitations of the system

NH_3 was selected as the target gas in this study because of its high emission rates from manure, regulatory significance, and substantial environmental impacts on air quality, ecosystem nitrogen loading, and indirect greenhouse gas formation (Wheeler et al., 2011). Additionally, NH_3 is known to be a "sticky" gas, as it is highly soluble in water and prone to adsorption on surfaces, which makes it challenging to measure accurately in chamber-based systems. Thus, a system that demonstrates high accuracy and recovery rates for NH_3 , may be expected to perform equally well, or even better, for less reactive and less soluble gases such as methane (CH_4) or nitrous

Table 2
Accuracy test results of the developed manure incubation system.

Chamber	Ammonia (NH_3) Concentration (ppm)			
	Replication 1	Replication 2	Replication 3	Mean $\pm$ SD ^a
Chamber 1	81.46	82.62	80.70	81.59 ± 0.79
Chamber 2	81.82	82.65	84.21	82.89 ± 0.99
Chamber 3	80.82	81.02	83.34	81.73 ± 1.14
Chamber 4	83.91	82.32	85.25	83.83 ± 1.20
Chamber 5	82.16	82.14	82.84	82.38 ± 0.33
Chamber 6	82.69	82.33	82.77	82.60 ± 0.19
Mean $\pm$ SD ^a	82.14 ± 1.07	82.14 ± 1.07	82.14 ± 1.07	82.50 ± 1.18

^a Standard deviation

Table 3

Precision test of the developed manure incubation system.

Chamber	Ammonia (NH ₃) Production Rate (mg/g manure/hour)					Mean±SD ^a
	Day 1	Day 2	Day 3	Day 4	Day 5	
Chamber 1	0.0034	0.0036	0.0037	0.0040	0.0037	0.0037 ± 0.0002
Chamber 2	0.0035	0.0036	0.0037	0.0036	0.0033	0.0036 ± 0.0002
Chamber 3	0.0038	0.0039	0.0039	0.0036	0.0034	0.0037 ± 0.0002
Chamber 4	0.0037	0.0040	0.0042	0.0041	0.0037	0.0039 ± 0.0002
Chamber 5	0.0033	0.0038	0.0042	0.0039	0.0035	0.0037 ± 0.0003
Chamber 6	0.0039	0.0040	0.0041	0.0041	0.0037	0.0040 ± 0.0001
Mean±SD ^a	0.0036 ± 0.0002	0.0038 ± 0.0002	0.0040 ± 0.0002	0.0039 ± 0.0002	0.0035 ± 0.0002	0.0038 ± 0.0003

^a Standard deviation

oxide (N₂O). Furthermore, methane and nitrous oxide are produced mainly under anaerobic conditions (Webb et al., 2011), whereas the current system operated under aerobic conditions, making their production negligible in this context. Therefore, NH₃ was intentionally chosen as the primary validation gas to ensure that the system could reliably handle the most demanding measurement scenario.

There are several gas emission measurement systems with controlled manure incubation conditions. The basic concept of these systems is to capture the gases produced by the incubation of manure in each chamber within a closed system, where experimental conditions such as ventilation, airflow rate, fermentation temperature, and the amount of manure can be controlled by researchers. However, since the flux chamber systems developed have been designed for specific research purposes, there are currently no standardized systems (Wheeler et al., 2011). For example, to compare gas production using the same manure under different incubation conditions such as temperature and airflow rate, a system that can be operated with different chamber conditions for a short period needs to be constructed. On the other hand, to evaluate the effects of manure incubation according to different feeds or additives, a system that can operate under the same experimental conditions for a long period in each chamber should be established. Therefore, the incubation system developed in this study, which shows high accuracy and precision results, can be considered a suitable system for manure research.

Moreover, the multi-chamber system developed in this study provides a practical alternative to highly automated but expensive existing model. The system developed by Wheeler et al. (2011) necessitates the use of individual MFCs and complex direct-to-analyzer air routing, which incurs increased manufacturing costs and difficulties in maintenance. The substitution of individual MFCs with a timer-controlled diaphragm sampling pump and an automated relay system to collect composite air samples into Tedlar bags over a 24-hour period has been demonstrated to effectively eliminate these structural bottlenecks. This indirect composite sampling design has been shown to enhance practicality while maintaining the precision required for reliable daily emission tracking.

Combining the results of the accuracy and precision tests, it can be concluded that the mixing of air within the chambers influenced the accuracy of the measurements. Considering the mixing effects observed in the experiments, it would be advisable to include a blank in each experiment. Despite the mixing effects, the precision of the measurements remained consistent across the chambers. These findings highlight the importance of considering the mixing effects and system design when interpreting and comparing NH₃ concentration results in such experimental setups. Further optimization of the system's design and airflow patterns may help minimize the mixing effects and improve the accuracy of the measurements.

Further studies should extend the validation of this system to other major greenhouse gases such as H₂S or methane, as recent work has demonstrated the feasibility of tracer-based and stationary methods for robust methane quantification from manure storage facilities (Vechi and Scheutz, 2023). Additionally, addressing the so-called N-gap by incorporating simultaneous measurements of dinitrogen (N₂) loss, alongside NH₃ and nitrous oxide, would provide a more comprehensive understanding of total nitrogen emissions from manure (Neysari et al., 2023).

Table 4

Effect of different crude protein levels in concentrate mix on fecal and urine characteristics.

Items	Treatment ^a				SEM	P-value		
	LCP	MLCP	MHCP	HCP		Mean	Linear	Quadratic
Feces								
Wet weight, kg/day	14.44	12.09	15.42	14.42	2.25	0.759	0.572	0.486
Total solid, kg/day	2.57	2.18	2.74	2.47	0.40	0.791	0.709	0.440
Total volatile solid, kg/day	2.20	1.86	2.36	2.13	0.33	0.765	0.674	0.431
Fecal nitrogen, g/day	59.22	48.41	64.52	57.50	8.71	0.635	0.593	0.345
Fecal sulfur, g/day	4.09	3.28	4.32	3.83	0.63	0.688	0.637	0.342
Urine								
Total volume, L/day	15.24	21.63	16.50	19.23	4.41	0.186	0.361	0.055
Urinary nitrogen, g/day	114.15	150.96	141.40	164.30	16.70	0.222	0.857	0.097
Urinary sulfur, g/day	33.05	42.88	30.78	53.26	10.30	0.166	0.304	0.051

^a LCP, Low crude protein (12%); MLCP, Middle-low crude protein (15.5%); MHCP, Middle-high crude protein (17.25%); HCP, High crude protein (19%).

3.2. Manure incubation

3.2.1. Daily NH_3 and H_2S production

There were no significant differences in fecal and urinary outputs due to the different crude protein levels in the concentrate mix ($P > 0.05$), nor were there significant differences in the nitrogen and sulfur excretion through feces and urine ($P > 0.05$; Table 4). The daily NH_3 and H_2S production from manure incubation showed varying patterns among the different treatments and incubation days (Fig. 2).

In general, NH_3 production increased from day 1–3, peaked on day 3, and then gradually decreased thereafter (Fig. 2A). On day 1, the production of NH_3 for the LCP, MLCP, MHCP, and HCP treatments were 0.0037 ± 0.003 , 0.0043 ± 0.003 , 0.0036 ± 0.003 , and 0.0055 ± 0.006 mg NH_3 /g manure/hour, respectively. By day 3, the production had risen to 0.0058 ± 0.001 , 0.0065 ± 0.001 , 0.0090 ± 0.003 , and 0.0130 ± 0.014 mg NH_3 /g manure/hour, and by day 10, the production had lowered to 0.0031 ± 0.001 , 0.0035 ± 0.001 , 0.0044 ± 0.002 , and 0.0065 ± 0.002 mg NH_3 /g manure/hour, respectively. On the other hand, the pattern of H_2S production was not as consistent as NH_3 (Fig. 2B). The production of H_2S also increased in incidence from day 1–3, peaking on day 3 or 4, and then followed a decreasing trend; however, the declining patterns were not consistent across treatments. On day 1, the production of H_2S for the LCP, MLCP, MHCP, and HCP treatments were 0.0019 ± 0.002 , 0.0015 ± 0.002 , 0.0011 ± 0.002 , and 0.0009 ± 0.001 μg H_2S /g manure/hour respectively. By day 3, the production showed a moderate increase to 0.0289 ± 0.018 , 0.0236 ± 0.019 , 0.0352 ± 0.012 , and 0.0285 ± 0.011 μg H_2S /g manure/hour. On day 10, the production of H_2S was 0.0019 ± 0.002 , 0.0042 ± 0.002 , 0.0012 ± 0.001 , and 0.0021 ± 0.001 μg H_2S /g manure/hour, respectively.

3.2.2. Cumulative NH_3 and H_2S production

On average over 10 days of incubation, the CP levels in the concentrate mix significantly affected the NH_3 emission rate of manure (Table 5). On average, the manure from the HCP treatment exhibited 1.91 times higher daily NH_3 production compared to the manure from the LCP and MLCP treatments ($P < 0.001$). Linear and quadratic trends were also observed with increasing CP levels ($P < 0.05$). This suggests that, although the total nitrogen concentration in manure was not altered, a higher CP level in the concentrate mix shifts the characteristics of manure toward emitting more NH_3 . Unlike NH_3 production, the average daily H_2S production rate of manure did not differ by treatment over 10 days. This indicates that the increase in CP level in the concentrate mix did not have a significant impact on H_2S production during the incubation period. The newly developed system can sensitively detect these specific changes in manure characteristics.

Despite significant differences in the NH_3 emission rate of manure among the treatments, the cumulative NH_3 emission did not differ by the CP level in the concentrate mix (Fig. 3A). The cumulative production of NH_3 seemed to increase linearly, and the cumulative NH_3 emission over 10 days of incubation was 1.61, 1.55, 2.15, and 2.38 g NH_3 /steer for the LCP, MLCP, MHCP, and HCP treatments, respectively, which were not significantly different ($P = 0.323$; Table 5). Rather than indicating a limitation, this apparent discrepancy highlights the system's robust capability to distinguish between two distinct metrics: "emission rate per unit mass" and "total cumulative emissions." While differences in cumulative emissions were obscured by substantial natural variations in total manure excretion volume among individual animals, the system effectively and sensitively captured the fundamental chemical shifts in

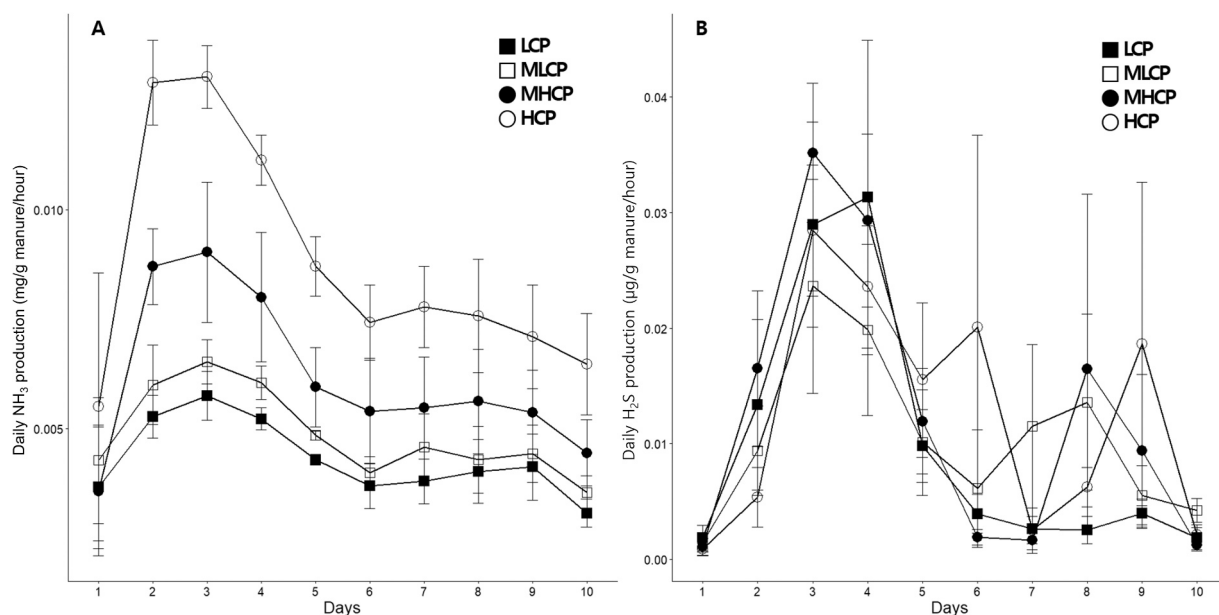
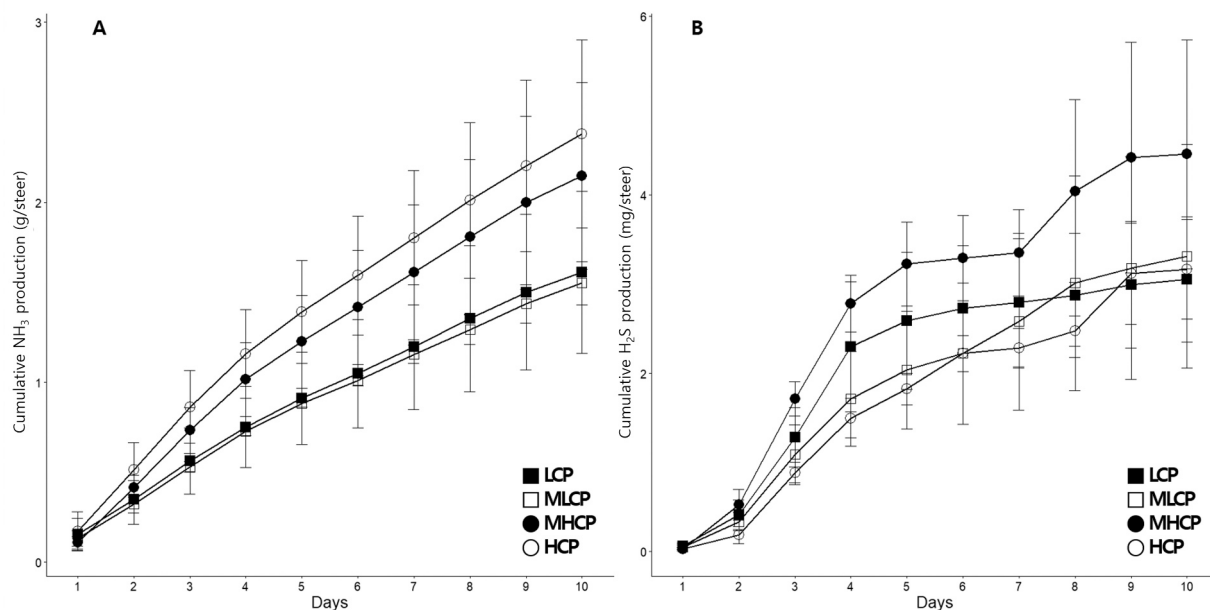


Fig. 2. Effect of different crude protein levels in concentrate mix on daily gas production. A, mg NH_3 /g manure/hour; B, μg H_2S /g manure/hour; ■, LCP; □, MLCP; ●, MHCP; ○, HCP.

Table 5Effect of different crude protein levels in concentrate mix on ammonia (NH₃) and hydrogen sulfide (H₂S) emissions from manure over 10 days.

Items	Treatment ^a				SEM	P-value		
	LCP	MLCP	MHCP	HCP		Mean	Linear	Quadratic
NH ₃ , mg/g manure/day	0.0043 ^a	0.0049 ^a	0.0062 ^{ab}	0.0088 ^b	0.00065	< 0.01	0.013	0.045
Cumulative NH ₃ , g/steer	1.61	1.55	2.15	2.38	0.393	0.323	0.245	0.812
H ₂ S, µg/g manure/day	0.0100	0.0106	0.0124	0.0125	0.00189	0.720	0.727	0.912
Cumulative H ₂ S, mg/steer	3.05	3.31	4.45	3.17	0.580	0.348	0.503	0.412

^{a-c}Means that do not have common superscripts significantly differ within the treatments ($P < 0.05$)^a LCP, Low crude protein (12%); MLCP, Middle-low crude protein (15.5%); MHCP, Middle-high crude protein (17.25%); HCP, High crude protein (19.0%).**Fig. 3.** Effect of different crude protein levels in concentrate mix on cumulative gas production. A, NH₃ g/steer; B, H₂S mg/steer; ■, LCP; □, MLCP; ●, MHCP; ○, HCP.

the manure substrate. Specifically, it detected an increased NH₃ emission rate, attributable to the HCP diet. The cumulative production of H₂S showed a logarithmic pattern (Fig. 3B). The cumulative H₂S emission over 10 days of incubation was 3.05, 3.31, 4.45, and 3.17 mg H₂S/steer for the LCP, MLCP, MHCP, and HCP treatments, respectively. the cumulative H₂S production also did not differ significantly according to treatment ($P = 0.348$; Table 5).

3.2.3. Interpretation and implications

From the results of manure characteristics and incubation, it can be inferred that while the CP level in the concentrate mix may influence the NH₃ emissions from manure, it does not significantly impact the concentration of total N and total S in the manure during the early fattening phase of steers. It is important to note that other factors such as environmental conditions, microbial activity, and sample variations could also influence the observed results. Nevertheless, the findings of this study indicate that feeding diets up to the MHCP level does not significantly increase daily NH₃ and H₂S production in early-fattening Hanwoo steers compared to the LCP diet.

Our research findings suggest that feeding steers with feeds of varying protein content can significantly influence the chemical characteristics of their manure, which in turn affects the emission of odor gases. Specifically, our results show no significant differences in NH₃ production between the MHCP and LCP diets. This indicates that maintaining a high nutritional level through the MHCP diet can be beneficial for the early-fattening Hanwoo steers without increasing the generation of odor gases. This is particularly important for livestock managers who aim to balance animal health and environmental sustainability. Overall, these results underscore the importance of considering the protein content of feed in relation to odor gas emissions from cattle manure. This study contributes to a deeper understanding of the relationship between cattle diet, manure composition, and gas emissions, guiding decision-making processes that prioritize both optimal cattle nutrition and environmental sustainability.

3.2.4. Practical applications and limitations

From an industrial perspective, the finding that the MHCP diet supports high nutritional performance without significantly increasing NH_3 or H_2S emissions offers a practical feeding strategy for early-fattening cattle. It is recommended that livestock producers employ such data-driven diet formulations to optimize protein supply while mitigating odor and greenhouse gas emissions, thereby aligning with environmental regulations and community concerns.

Moreover, the present study demonstrates that the simplified multi-chamber manure incubation system can sensitively detect subtle changes in emission profiles attributable to dietary modifications, providing a reliable platform for controlled, repeatable gas emission assessments. This system offers researchers an accessible and precise alternative to more complex and costly chamber designs, enabling broader adoption in nutrition–environment interaction studies. Furthermore, the system could be integrated into feed evaluation protocols or environmental monitoring programs on farms, thereby supporting evidence-based decisions for sustainable manure and odor management in the cattle industry.

While this system provides a robust platform for comparative assessments, it is important to acknowledge its limitations regarding extrapolation to field conditions. The standardized parameters employed in this study, including a constant temperature of 20°C and a ventilation rate of 2 L/min, were deliberately selected to isolate and compare the precise effects of dietary treatments. However, it must be noted that these constants may not fully represent fluctuating seasonal barn temperatures or dynamic farm ventilation rates. Furthermore, while the 10-day incubation period effectively captures the initial peak and dynamics of gas production during early fermentation, it may not reflect the entire complex chemical evolution of a full manure storage cycle. It is imperative that future studies operate this system under varied environmental variables. Only then will it be possible to fully bridge the gap between controlled dietary evaluations and real-world farm applications.

4. Conclusion

This study developed and validated a simplified multi-chamber manure incubation system that replaces complex real-time air routing with a practical 24-hour composite sampling approach. The system showed high measurement accuracy (95.7% recovery) and precision (2.6% coefficient of variation) for NH_3 , with consistent performance across chambers. Application of the system demonstrated that dietary CP level affected daily NH_3 emission rates, with the high-protein (HCP) diet producing significantly greater emissions than low (LCP) and medium-low (MLCP) CP diets, while NH_3 emissions from the medium-high (MHCP) and LCP diets were comparable. H_2S production patterns were more variable and showed no significant differences among treatments.

These results suggest that dietary CP can influence NH_3 emission rates without altering total nitrogen content in manure, and that the MHCP diet may maintain high nutritional value while limiting odor gas emissions in early-fattening steers. The system has been developed to offer a cost-effective and standardized tool for evaluating manure management strategies. This simplified system bridges the gap between complex laboratory instrumentation and practical nutritional research. Future research should extend the validation to other gases such as methane and H_2S under varying environmental variables. This would further inform sustainable livestock management that balance productivity with environmental sustainability.

CRedit authorship contribution statement

Kyewon Kang: Investigation, Formal analysis. **Seoyoung Jeon:** Investigation, Data curation. **Sinyong Jeong:** Investigation, Formal analysis. **Mingyung Lee:** Investigation, Formal analysis, Data curation. **Hyunjin Cho:** Investigation, Formal analysis, Data curation. **Kyungmo Kang:** Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Seongwon Seo:** Writing – review & editing, Visualization, Supervision, Methodology, Data curation, Conceptualization. **Heekwon Ahn:** Writing – review & editing, Visualization, Supervision, Project administration, Data curation, Conceptualization. **Jinho Shin:** Investigation, Formal analysis. **Hamin Kang:** Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare no competing interests.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.anifeedsci.2026.116816](https://doi.org/10.1016/j.anifeedsci.2026.116816).

Data availability

The datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

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